

WEEK 7 IMPLEMENTATION REPORT

Course Progression Architecture & Foundational React 19 SPA Setup

Project Phase:	Week 7 Implementation	Status:	COMPLETED (80/80 Tests Passed)
Author:	Engineering Architecture Team	Date:	September 16, 2026
Technology Stack:	Spring Boot 3.2, React 19, TypeScript, Vite, Tailwind v4	Test Suite:	80 Backend Tests + Vite Build 0 Errors

1. Executive Summary & Objectives

Week 7 accomplishes two major milestones for LearnPulse LMS: extending the backend domain model to support mastery-based **Course Progression Architecture** (Course -> Chapters -> Chapter Quizzes -> Progression -> Final Quiz) and establishing the foundational **React 19 Single Page Application (SPA)** infrastructure with TypeScript, Vite, Tailwind CSS v4, Axios, React Router v7, and React Hook Form. Server-side progression rules guarantee that Chapter 1 is unlocked by default, passing Chapter N quiz with $\geq 80\%$ score unlocks Chapter N+1, passing **ALL chapters belonging to the Subject/Course** unlocks the Final Course Quiz, and scoring $\geq 75\%$ on the Final Quiz marks the course completed.

2. Progression Architecture Hierarchy

The domain extends the existing Subject entity container without creating duplicate structures (diagram shows an illustrative N-chapter flow):

Subject (Course Container: e.g. JAVA101)
■■■ Chapter 1: Basics -> Chapter Quiz 1 (Passing: 80%) [UNLOCKED BY DEFAULT]
■■■ Chapter 2..N: Concepts -> Chapter Quiz N (Passing: 80%) [LOCKED until N-1 Passed]
■■■ Final Course Assessment (Passing: 75%) [LOCKED until ALL Chapters Passed]
■■■ Status: COMPLETED upon scoring $\geq 75\%$

3. Extended Domain Enums & Entities

Component	Type / Target	Details & Values	Purpose / Constraints
QuizType	Enum	CHAPTER_QUIZ, FINAL_COURSE_QUIZ	Discriminator for progression rules
QuizStatus	Enum	DRAFT, PUBLISHED	Teacher publishing workflow control
QuestionSource	Enum	TEACHER_MANUAL, TEACHER_UPLOADED, AI_GENERATED	Supported question source architecture
Quiz Entity	JPA Entity	passingScorePercentage (80.0 / 75.0), status, quizType	Linked to Subject and Chapter
Question Entity	JPA Entity	teacherPriority, customInstruction, source	Extension metadata for question sources

4. React 19 Frontend SPA Infrastructure

Layer / Module	Technology / File	Implementation Details & Features
HTTP Client & Auth Interceptor	src/api/axios.ts	Axios instance with automatic Bearer token injection & 401 refresh token handler
Authentication State Context	src/context/AuthContext.tsx	Global AuthProvider storing user state, login, register, logout, and token persistence
Protected Route Guard	src/components/ProtectedRoute.tsx	RBAC authorization wrapper protecting STUDENT, TEACHER, and ADMIN routes
Registration Security	src/pages/auth/RegisterPage.tsx	Public registration restricted strictly to STUDENT and TEACHER roles (ADMIN excluded)
Application Pages	src/pages/*	Built Login, Register, Student Dashboard/Practice/Results/Profile, Teacher Dashboard/Questions, Admin Dashboard/Users

5. Automated Testing & Verification Results

The complete backend Maven integration test suite runs **80 automated integration tests** with **0 failures**. Frontend TypeScript compilation (tsc -b) and Vite production bundling (vite build) completed with **0 errors**.

Test Class Name	Tests	Status	Coverage Area
CourseProgressionIntegrationTest	10	10 PASSED	Single & multi-chapter progression locking/unlocking
AssessmentEngineIntegrationTest	39	39 PASSED	Quiz delivery, answer reveals, server grading
SecurityRbacIntegrationTest	7	7 PASSED	JWT authentication, RBAC authorization, registration
DocumentIngestionIntegrationTest	9	9 PASSED	PDF/DOC/DOCX upload & Tika extraction
StudentTeacherProfile & Academic	10	10 PASSED	Profiles, Subject & Chapter hierarchy
JwtProvider & Application Context	5	5 PASSED	Token generation, status endpoints, app context
TOTAL PROJECT SUITE	80	80 PASSED	BUILD SUCCESS (Frontend Build 0 Errors)

6. Verification & Mentor Sign-Off

Verification Confirmed: All Week 7 Course Progression Architecture extensions and foundational React 19 Frontend SPA components have been fully verified with 80 passing backend tests and a clean production Vite build. Final Course Quiz unlocks only after ALL chapters have been passed, QuestionSource supports TEACHER_MANUAL, TEACHER_UPLOADED, and AI_GENERATED, and public user registration excludes ADMIN privileges.